V-Droid: Advancing Mobile GUI Agent Through Generative Verifiers

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Paper version: arXiv 2503.15937v5 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *the highest SR achieved by existing agents is only 54.3%, significantly lower than the human performance of 80%*

LLM **generators** decode long chains per step — SR stalls at 54.3%, Agent-S2 25 s+ per step.

Why verification fits mobile

1. **Gen–verification gap.** Judging is easier than generating.
2. **Small action space.** ≈20 interactive elements per page — enumerable.
3. **Raw verifiers fail.** Untrained 8B: 0% SR; process labels absent.

Evaluation setup

AndroidWorld · 116 | AndroidLab · 138 | MobileAgentBench · 100 | Llama-3.1-8B (4-bit) | 110K training pairs

- **All online**; AndroidLab + MobileAgentBench apps held out (AndroidWorld in-domain).
- **Latency:** 20 sampled tasks; baseline SRs from their papers.

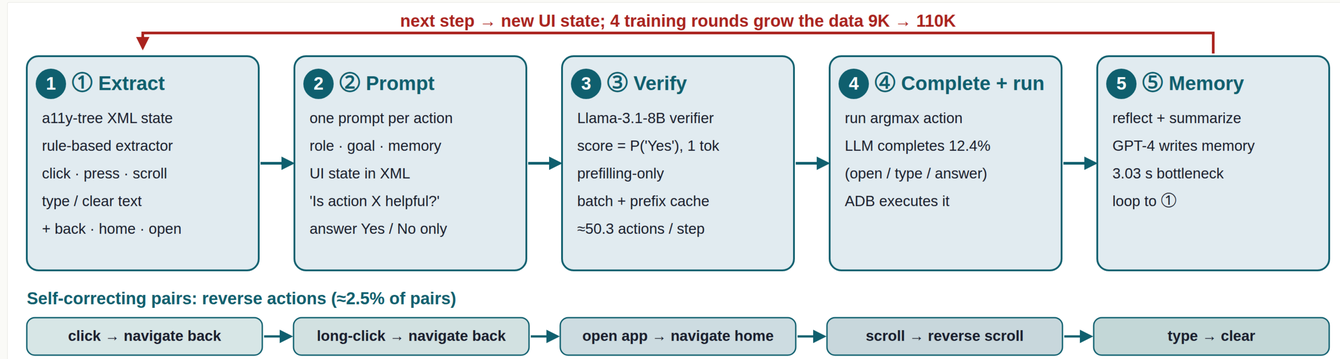
Our read (CUA Radar)

65	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval ×3; held-out apps (2/3); strong baselines; latency; ablation; AUC; code.
- **Flags:** AndroidWorld in-domain; baselines not rerun; no variance; a11y list incomplete.
- Pattern K02. Authors' numbers, not rerun.

Method: score extracted actions, don't generate them

- **Sees / acts:** a11y-tree XML + working memory in; discrete actions (click, long-press, scroll, type/clear, nav, open app) via ADB.
- **Verifier:** Llama-3.1-8B scores every candidate by P('Yes') — one prefill token, batched, prefix-cached; 0.44 s for ≈50.3 actions.
- **Training:** P³ preference loss on positive-vs-negative pairs per step + reverse-action self-correction; 4 annotation rounds, 9K → 110K pairs.



Our redraw (not the paper's figure): one step of the V-Droid loop, plus the reverse-action map behind self-correcting training.

$$L_\kappa(t) = -\mathbb{E}\left[\log \sigma\left(\tau(\alpha_\kappa^y(t)) - \tau(\alpha_\kappa^n(t))\right)\right]$$

P³ loss (Eq. 3): the labeled action must outscore every other candidate at each step

Key results: AndroidWorld success rate (%)

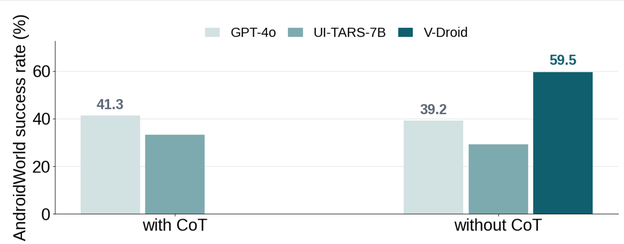


Table 2, redrawn: generator agents with / without CoT vs V-Droid.

59.5% AndroidWorld · 38.3% AndroidLab · 49% MobileAgentBench

SO WHAT → *+5.2 / +2.1 / +9.0 pts over prior best; V-Droid needs no CoT.*

Same 8B, other jobs (AndroidWorld SR, %)

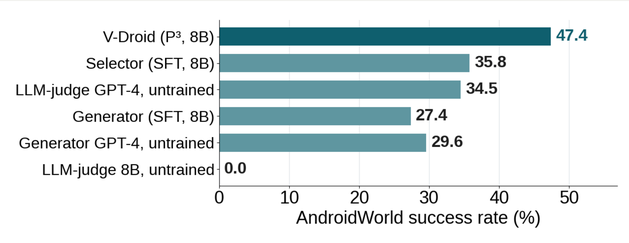


Table 3, redrawn: architecture and training alternatives, all trained on the same 1.1K task steps.

SO WHAT → *The gain is the paradigm plus P³: the same Llama-3.1-8B trained as selector or generator reaches only 35.8 / 27.4, and untrained judges 34.5 / 0 — verification works only after preference training.*

Headline Numbers

59.5%
AndroidWorld SR
prior best 54.3% · human 80%

- **38.3%** AndroidLab SR
- **49%** MobileAgentBench
- **0.74 s** per decision

The paper's own controls

CONTROL (ANDROIDWORLD)	V-DROID	ALTERNATIVES
− self-correct pairs	59.5	52.2 without them
Rule / history memory	59.5	46.1 / 40.0
Training data 9K → 110K	15.0 → 59.5	17.0 → 49.0
Latency breakdown (s)	4.3 total	3.03 mem · 0.44 verify

Same base, same data: P³ 47.4 vs selector 35.8 / gen 27.4.

Takeaway

An 8B verifier scores extracted candidates rather than generating them: 59.5% AndroidWorld vs 80% human, 4.3 s/step, P³ on 110K pairs.

9K → 110K pairs, 4 rounds | ≈50.3 actions/step | ≈1 min/task human fix | 16 GB GPU incl. KV cache

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Paper | Code